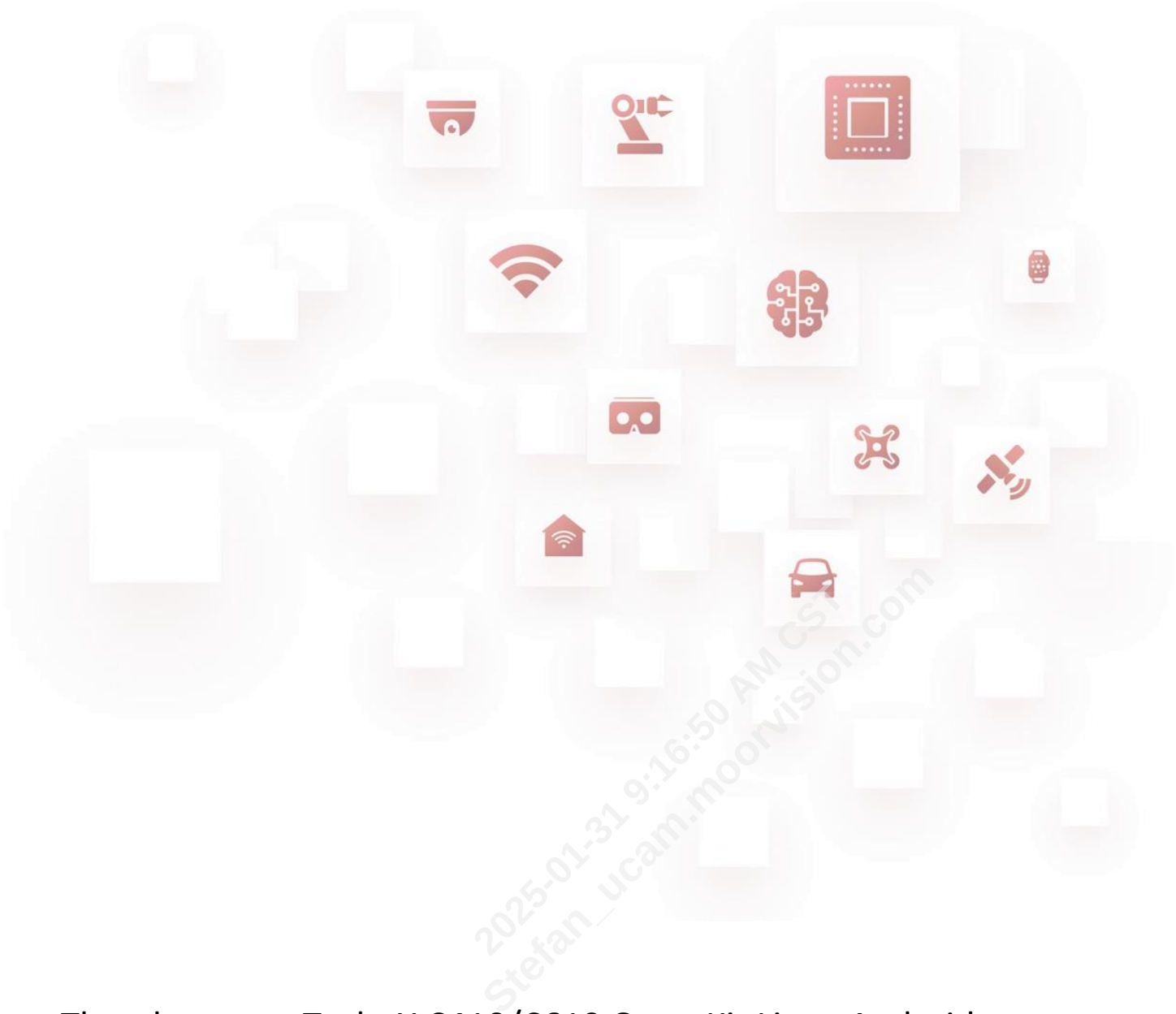
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Thundercomm TurboX C410/C610 Open Kit Linux Android

Software Release Notes

Rev. V1.0
July 26, 2022

DN: tc-gla-2410
SDK Version: LA1.0.CS.r002005.1

Revision History

Revision	Date	Description
1.0	July 26, 2022	Initial release.

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About This Document

- Illustrations in this documentation might look different from your product.
- Depending on the model, some optional accessories, features, and software programs might not be available on your device.
- Depending on the version of operating systems and programs, some user interface instructions might not be applicable to your device.
- Documentation content is subject to change without notice. Thundercomm makes constant improvements on the documentation of the products, including this guidebook.
- Function declarations, function names, type declarations, attributes, and code samples appear in a different format, for example, `cp armcc armcpp`.
- Code variables appear in angle brackets, for example, `<number>`.
- Button, tool, and key names appear in bold font, for example, click **Save** or press **Enter**.
- Commands to be entered appear in a different font; on the host computer use $ as shell prompt, while on the target device use # as shell prompt, for example,

```
$ adb devices
# logcat
```

- Part of the code that does not contain instructions appear in a different format, for example,

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
```

- Folders and files are formatted in italic, for example, *turbox_flash_flat.sh*.

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Chapter 1. Introduction

This document provides information on how to obtain, build, and program QCS410/610 LA (Linux Android) SDK (Software Development Kit) into the TurboX C410/C610 Open Kit.

➲ **NOTE:** "TurboX" referred to herein is the English text of our registered trademark TURBOX.

The Linux Android Software Development Kit supports the Android operating system on the Thundercomm TurboX. The goal of the LA SDK is to provide a simple and flexible development platform to rapidly build IoT products with the QTI chipsets. The LA SDK provides the build framework, software components, tools, and sample applications to evaluate the platform and build software images for the target product.

➲ **NOTE:** The LA Software can run on the 4GB DDR TurboX C610 platform with best performance.

Table 1-1. TurboX C410/C610 platform specifications

SOC	SKUx		Standard SKU[1)]	OS	Thundercomm Product No. (TPN)	Model
QCS610	SKU1	2+16GB	Yes	LE/LA[2)]	C610-D21-369PWXX C610-D21-LCXPW00 (LE Specified) C610-D21-ACXPW00 (LA Specified)	TurboX C610
	SKU2	2+64GB	No	LE/LA[2)]	C610-D26-369PWXX C610-D26-LCXPW00 (LE Specified) C610-D26-ACXPW00 (LA Specified)	TurboX C610
	SKU3	4+64GB	Yes	LE/LA	C610-D46-369PWXX C610-D46-LCXPW00 (LE Specified) C610-D46-ACXPW00 (LA Specified)	TurboX C610
QCS410	SKU4	2+16GB	Yes	LE/LA[2)]	C410-D21-369PWXX C410-D21-LCXPW00 (LE Specified) C410-D21-ACXPW00 (LA Specified)	TurboX C410
QCS410	SKU5	2+64GB	No	LE/LA[2)]	C410-D26-369PWXX C410-D26-LCXPW00 (LE Specified) C410-D26-ACXPW00 (LA Specified)	TurboX C410
QCS410	SKU7	4+64GB	No	LE/LA	C410-D46-369PWXX C410-D46-LCXPW00(LE Specified) C410-D46-ACXPW00(LA Specified)	TurboX C410
QCS410	SKU8	2+8GB	No	LE	C410-D28-369PWXX C410-D28-LCXPW00 (LE Specified)	TurboX C410

➲ **NOTE:**

[1)] indicates standard SKU series. For standard SKU series, retail stock is available; as to non-standard SKU series, in case of any customer requirement, specific orders need to be placed accordingly.

[2)] indicates that when the memory is less than 2GB, Android OS may run out of memory; in this case, it is necessary to tailor the system as required to stabilize the system.

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Chapter 2. System Requirements

Table 2-1 lists the hardware and software required to install and run the LA SDK.

Table 2-1. Hardware and software required to install and run the LA SDK

Hardware / Software Requirements	Source / Vendor	Function
Standalone hardware system minimum requirement: • 16 GB RAM • Quad core CPU ➲ **NOTE**: System with lower hardware specifications will increase compilation time. About an hour is the compilation time for a system with 16 GB RAM and Intel i7-2600 @3.4 GHz.	–	SDK build machine
Ubuntu 16.04/18.04 LTS Linux distribution for 64 bit architecture	Ubuntu Community/ Canonical, Ltd.	SDK build host OS
Repository	SDK Open Source Project	SDK source management tool
Python 2.7.12 with sqlite3 module	Python.org	Building subsystem

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Chapter 3. Set up the Development Environment

This chapter introduces the setup of a Linux development environment on an Ubuntu host machine.

Prerequisites: Ensure your host meets the following requirements:

- Host machine Linux distro – 16.04/18.04 LTS
- Minimum 300 GB free disk space
- Tar: V1.27 or newer
- Privileges to install utilities
- Install the host packages

3.1. Ubuntu 16.04/18.04 (64-bit) installation

Step 1. Install ubuntu 16.04/18.04 on the computer following the instructions at: *http://www.ubuntu.com/download/desktop/install-ubuntu-desktop*.

Step 2. To update software, execute the following command.

```
$ sudo apt-get update
```

3.2. Tool and installation

This section provides instructions for installing **ADB** (Android Debug Bridge) (V1.0.36 or later), **Fastboot**, and USB drivers to your host computer.

The **Fastboot** is a tool used to flash images onto the board by communicating with the boot loader.

The **ADB** is a tool installed on the host computer to access the device Unix shell, then help to push, modify, debug, and tweak device system files from the host computer.

3.2.1. ADB and Fastboot tools installation

To install **ADB** and **Fastboot**, execute the following commands:

```
$ sudo apt-get install android-tools-adb
$ sudo apt-get install android-tools-fastboot
```

3.2.2. USB driver installation and configuration

To install and configure USB drivers, follow the steps below:

Step 1. Set up the udev rules on a Linux computer.

1) Login as **root/sudo**.

```
$ cd /
```

2) Navigate to the directory */etc/udev/rules.d/*.

```
$ cd /etc/udev/rules.d/
```

3) Create the file *99-android.rules.*

```
$ sudo touch 99-android.rules
```

4) Enter the following command get edit permission to file *99-android.rules*.

```
$ sudo chmod 777 /etc/udev/rules.d/99-android.rules
```

5) Enter *99-android.rules* by running the following command.

```
$ vim 99-android.rules
```

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6) Input `i` to enter edit mode and add the following content to the file.

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
SUBSYSTEM=="usb", ATTR{idVendor}=="05c6", MODE="0777", GROUP="adm"
```

7) Press **ESC** key to exit the edit mode.

8) Enter `:wq` to save the change and exit.

Step 2. Restart the udev service by executing the following commands.

```
$ sudo chmod a+r /etc/udev/rules.d/99-android.rules
$ sudo service udev restart
```

Step 3. Connect the board to your computer with an USB cable, and power on the target board to start the system. Wait for several seconds then input `$ adb devices`. The device is correctly detected with the following information displayed on your screen.

```
$ adb devices
List of devices attached
xxxxxxxx        device
```

➲ **NOTE:** Avoid connecting the device to the PC with a USB hub. Connect the device to the PC directly via a USB cable.

3.3. Required building environment

Follow the steps below to configure the building environment.

3.3.1. Java 8

Step 1. Check the version. Run the following commands to ensure Java 8 is installed.

```
$ java -version
openjdk version "1.8.0_131"
OpenJDK Runtime Environment (build 1.8.0_131-8u131-b11-0ubuntu1.16.04.2-b11)
OpenJDK 64-Bit Server VM (build 25.131-b11, mixed mode)
```

Step 2. Version update.

With a 1.7 version preinstalled, run the following commands to upgrade it to version 1.8.

```
$ sudo apt-get update
$ sudo apt-get install openjdk-8-jdk
```

3.3.2. Linux package dependencies

Step 1. Install the dependency libraries to the host computer.

```
$ sudo apt-get install git-core gnupg flex bison build-essential zip curl \
zlib1g-dev gcc-multilib g++-multilib libc6-dev-i386 lib32ncurses5-dev \
x11proto-core-dev libx11-dev lib32z1-dev libgl1-mesa-dev libxml2-utils xsltproc \
unzip fontconfig libxml-simple-perl libssl-dev
```

Step 2. Set **bash** as default shell tool.

```
$ sudo rm /bin/sh
$ sudo ln -sf /bin/bash /bin/sh
```

➲ **NOTE**: For more information on building environment dependencies, refer to AOSP documents at:

https://source.android.com/setup/build/initializing.

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3.3.3. Install repo

Run the following command to install repo.

```
$ sudo apt-get install repo
```

3.3.4. Configure git global variable

Configure user name and email.

```
$ git config --global user.email "<User Email>"
$ git config --global user.name "<User Name>"
```

➲ **NOTE:** Replace the `<User Email>` and `<User Name>` with the email address and user name you have registered at Thundercomm website.

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Chapter 4. Get Started with SDK

4.1. SDK information

Table 4-1 presents the LA SDK information, including the SDK version.

Table 4-1. LA SDK information

SDK Version	Baseline	Baseline Distribution	OS Version
LA1.0.CS.r002005.1	QCS610.LA.1.0	Post-CS 0.0.054.3	Android 10.00

4.2. SDK block diagram

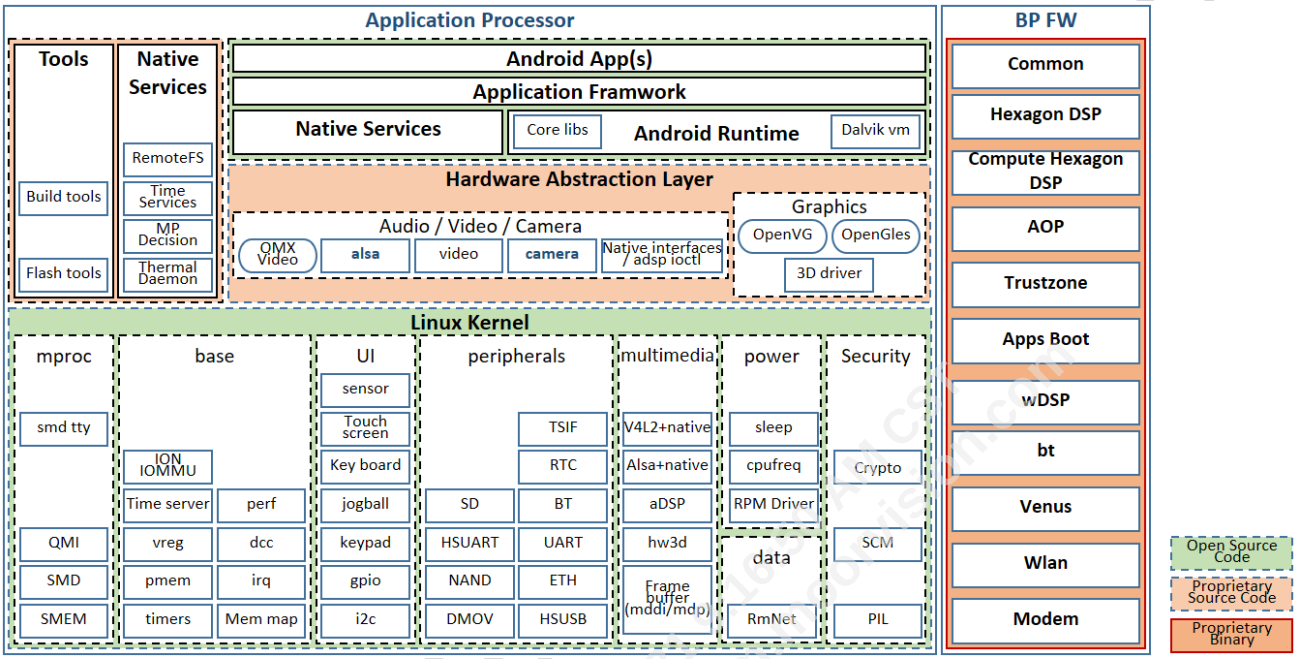


Figure 4-1. Block Diagram

4.3. SDK source tree structure

```
SDK.Turbox_C610.LA1.0/
├── about.html -> common/about.html            SDK version info(Product,
Distribution,Components, …)
├── contents.xml -> common/contents.xml                image contents file
├── common/config/emmc/partition.xml                        eMMC partition file
├── LINUX
│   └── android
│       ├── bionic
│       ├── bootable/bootloader/edk2                        bootloader sources
│       ├── bootable/recovery                          recovery sources
│       ├── build              basic config files for development kit, such as files for
system build rules and generis
│       ├── dalvik
│       ├── developers                  developer directory
│       ├── development             infrastructure-related files for Android application
development
│       └── device                  device related config
│           └── qcom/sm6150              QCS610 device-related config directory
│       ├── external                     files related to open-source development kit
│       ├── frameworks                   source code under Android frameworks
│       ├── hardware/qcom                code for QCS610 hardware abstraction layer
 (bootctrl, Display, gps, media, vr, …)
```

```
│       ├── kernel
│       │   └── msm-4.14                              kernel sources
│       │       ├── arch/arm64/boot/dts/qcom              DT source files
│       │       └──arch/arm64/configs/vendor              kernel config files
│       ├── libcore                                   core library files
│       ├── libnativehelper                           dynamic library, which is the
basis to JNI library
│       ├── Makefile                                  global Makefile to define build
rules
│       ├── packages                                  application packages
│       ├── pdk                                       plug development kit
│       ├── platform_testing                          platform testing files
│       ├── prebuilts                                 prebuild resources under x86 and
arm framework
│       ├── sdk                                       sdk and simulator
│       ├── system                                    underlying file system, application,
and component
│       ├── test                                      Android Vendor testing framework
│       ├── toolchain                                 tool chain files
│       ├── tools                                     tools files
│       └── vendor                                    QCOM private repository
│           └── qcom
│               ├── opensource
│               └── proprietary
├── turbox_build.sh -> turbox/tools/build_script/common/turbox_build.sh       SDK build
file.
├── turbox_flash.sh -> turbox/tools/build_script/device/turbox_flash.sh       Fastboot
flash tool
├── turbox
│   └── tools
│       ├── build_script                              SDK build script(android build,
zip, …)
│       └── flash_tools                               Flash tools files(qdl, …)
├── adsp_proc             aDSP image is released as a binary and no build compilation is
needed
├── aop_proc              AOP image is released as a binary and no build compilation is
needed
├── boot_images           BOOT image is released as a binary and no build compilation is
needed
├── btfm_proc             BTFM image is released as a binary and no build compilation is
needed
├── btfm_proc_CMC         BTFM_CMC image is released as a binary and no build compilation
is needed
├── cdsp_proc             cDSP image is released as a binary and no build compilation is
needed
├── modem_proc            MODEM image is released as a binary and no build compilation is
needed
├── trustzone_images      TZ image is released as a binary and no build compilation is
needed
├── venus_proc            VIDEO image is released as a binary and no build compilation is
needed
├── wdsp_proc             wDSP image is released as a binary and no build compilation is
needed.
└── wlan_proc             WLAN image is released as a binary and no build compilation is
needed.
```

4.4. Download SDK

Follow the steps below to activate an account and download the SDK code.

4.4.1. Account activation

Add the account/password information to *$HOME/.netrc* by running the following command.

```
$ echo "machine partner.thundercomm.com login <User Email> password <password>" >> ~/.netrc
```

➲ **NOTE:** Please use the email/password you have registered at www.thundercomm.com website to replace `<User Email>` and `<password>` in the above command.

4.4.2. Download the SDK code

Follow the steps below to download the SDK code.

Step 1. Initialize the repository.

```
$ repo init -u https://partner.thundercomm.com/sdkgit/a/manifest -b
SDK.Turbox_C610.LA1.0 -m turbox-c610-la1.0-dev.release.CS.r002005.1.xml --repo-
url=https://partner.thundercomm.com/sdkgit/a/tools/repo --repo-branch=stable --no-
repo-verify
```

Step 2. Synchronize the source code.

```
$ repo sync -c -j4 --no-tags
```

4.4.3. SDK version update

If you have downloaded the old-version SDK, there are two ways to download the new SDK fast.

Option 1. Upgrade the local old folder to a new release

Navigate to the old local code folder, you can run the repo init command of new release as specified in *4.4.2. Download the SDK code*, and then run `repo sync -j4 -c --no-tags` to receive the new code.

Option 2. Download the new release to a new folder(recommended)

Create a new folder, and run `repo init` with `-reference` value.

For example:

```
$ repo init -u https://partner.thundercomm.com/sdkgit/a/manifest -b
SDK.Turbox_C610.LA1.0 -m turbox-c610-la1.0-dev.release.CS.r002005.1.xml --repo-
url=https://partner.thundercomm.com/sdkgit/a/tools/repo --repo-branch=stable --
no-repo-verify --reference=<old_sdk_code_path>
$ repo sync -j4 -c --no-tags
```

➲ **NOTE:** In `--reference=<old_sdk_code_path>`, the `old_sdk_code_path` is the absolute path of your old sdk folder. For example: `--reference=/home/host/Turbox_C610.LA1.r001003`.

4.5. Build images

To compile code, run the following script file in the root directory of source code for building userdebug version.

```
$ ./turbox_build.sh -a -l -v userdebug
```

4.6. Pack ROM package

Step 1. Run the following command to update common info for userdebug version.

```
$ ./turbox_build.sh -u -v userdebug
```

➲ **NOTE**: The command executes successfully with the following log displaying on shell.

```
- update_common_info.py:======  UPDATE COMMON INFO COMPLETE=======
```

Step 2. Run the following command to pack the ROM package. You can find the ROM package at *turbox/output*.

- Pack **Fastboot** image

```
$ ./turbox_build.sh --zip_fastboot_build -l -v userdebug
```

- Pack **Flat** image

```
$ ./turbox_build.sh --zip_flat_build -l -v userdebug
```

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➲ **NOTE:** Customers can also get ROM package provided by Thundercomm from:

- *FastbootBuild_TurboX_C610_xx.xx_LA1.0.R.userdebug.CS.r002005.1.zip*

  MD5: d669318d5e44492ee0408cc89bef345a

- *FlatBuild_TurboX_C610_xx.xx_LA1.0.R.userdebug.CS.r002005.1.zip*

  MD5: e7fffddb9581e0a5d0798fec00ce6f03

4.7. Flash images to board

➲ **NOTE**: If the device fails to boot up after image flash with **Fastboot**, try to flash the **FlatBuild** image again with *turbox_flash_flat.sh or TurboX Easy Flash*.

4.7.1. Flash images on Windows

You can flash images on Windows by using TurboX Easy Flash, which is a tool created by Thundercomm to help SOM developers easily flash device image, backup and restore data, etc.

For tool acquisition and operation procedures, refer to *TurboX Easy Flash User Guide*.

4.7.2. Flash images on Linux

- **Flash images on Linux with Fastboot**

  Step 1. Enter the **Fastboot** mode.

  Option 1. Enter the following commands when the device is in **ADB** mode.

```
$ adb root
$ adb reboot bootloader
```

  Option 2. Power off the device, hold the **Volume down (VOL-)** button and press the **Power on** button, then release the **Volume down (VOL-)** button about 5 seconds later.

  ➲ **NOTE**:

  To check if the device has entered the **Fastboot** mode, run the following command:

```
$ sudo fastboot devices
```

  Successful with the following output appearing on the shell.

```
21f10525  fastboot
```

  Step 2. Flash images.

  Execute the commands below to flash images to the board. Replace `<folder name of image package>` with the actual image package name of your product.

```
$ cd <folder name of image package>
$ sudo ./turbox_flash.sh --all -r -f
```

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➲ **NOTE**: Enter `./turbox_flash.sh -h` to acquire detailed usage.

```
$ ./turbox_flash.sh -h
Usage:
  bash ./turbox_flash.sh [options] [options]
  Images except gpt,userdata,persist will be flashed if no options input

Description:
  flash images with fasboot

Options:
  -h, --help   display this help message
  -v, --variant <debug/perf/user>    set the build variant.(default: debug)
  -f, --force  forcibly flash gpt,persist and userdata without confirmation.
               User data and calibration data(BT/WIFI MAC address, etc..) will
be ereased.
  -r, --reboot      reboot the device automatically after flashing completes.
  -b, --nonhlos     flash all bp images(xbl,aop,ftfm,tz,dsp,modem,pmic).
  -a, --hlos        flash all ap
images(aboot,bootimage,recovery,system,vendor,cache,mdtp,userdata,persist).
  --all        flash all images

Example:
  ./turbox_flash.sh
      Flash the images except gpt,userdata,persist.

  ./turbox_flash.sh --force
      Flash all images.

  ./turbox_flash.sh -hlos
  Flash all ap images.
```

- **Flash images on Linux with *Turbox_flash_flat.sh* tool**

  Step 1.Download the flashing tool: turbox_flash_flat.sh.

  Step 2.Prepare the FlatBuild images.

  Check if the images are ready, then copy the *turbox_flash_flat.sh* to image folder.

  Replace `<folder name of image package>` with the actual image package name of your product.

```
$ cp turbox_flash_flat.sh <folder name of image package>
$ cd <folder name of image package>
$ chmod +x turbox_flash_flat.sh
```

  Step 3.Enter 9008 Mode.

  1) Power off the device by disconnecting the power cable and USB cable.

  2) Connect the power cable.

  3) Hold the **Force USB Boot** button.

  4) Press the **Power on** button and wait for about 3 seconds.

  5) Release the **Force USB Boot** button and **Power on** button.

  6) Connect the Type-C USB cable to an Ubuntu computer.

  To check if your device has entered the **9008 Mode**, run the following command:

```
$ lsusb
……
Bus 001 Device 042: ID 05c6:9008 Qualcomm, Inc. Gobi Wireless Modem (QDL mode)
…… .
```

Step 4.Start to flash the FlatBuild images.

Run the following command under the image folder.

```
$ sudo ./turbox_flash_flat.sh -x patch0.xml,rawprogram_unsparse0.xml -
f ./prog_firehose_ddr.elf -i ./ -s eMMC
```

If the flashing process is successful, the following log will be printed.

```
15:41:09: INFO:                _              (done)
15:41:09: INFO:       | |
15:41:09: INFO:     __| | ___  _ __   ___
15:41:09: INFO:   / _` |/ _ \| '_ \ / _ \
15:41:09: INFO: | (_| | (_) | | | |  __/
15:41:09: INFO:  \__,_|\___/|_| |_|\___|
15:41:09: INFO: {All Finished Successfully}

15:41:09: INFO: Overall to target 201.617 seconds (20.88 MBps)
15:41:09: INFO: {percent files transferred 100.00%}
```

Step 5.Reboot the device.

1) Power off the device by disconnecting the power cable and USB cable.

2) Connect the power cable to the board.

3) Press the **Power on** button.

4) Connect the USB 3.1 Gen 1 Type-C cable and check the adb device with the following command.

```
$ adb devices
List of devices attached
b237534        device
```

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Chapter 5. Features

This chapter introduces the software features supported by the DK.

5.1. Presented features

This section describes the key software features that have been included in earlier versions.

- Kernel
  - Linux Kernel 4.14
  - 64-bit only
- OS
  - Android 10
  - F2FS + FBE
  - Dynamic partition
  - Fastboot support
  - Boot image header version 1
  - Camera 2.0 (HAL 3.4)
  - Keymaster 4.0
  - Vulkan 1.1
  - FOTA
  - Incremental OTA
- CV
  - FastCV
- Security
  - Secure boot
  - Secure video encoding
  - Secure camera
  - Secure DSP
- Connectivity
  - WLAN 802.11ac
  - USB 3.1 Gen 1
  - Bluetooth 5.0
- Storage
  - eMMC
  - Dual rank2 channel LPDDR4x memory support
- Multimedia
  - Secure camera
  - YUV + Bayer capture
  - sHDR v3.8
  - sHDR tuning through Chromatix
  - LDC
  - TNR (MCTF)

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5.2. New features

This section describes the new key software features that will be included in this SDK version.

- Up to 8 GB memory map (QCS610)

5.3. On-going features

None.

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Chapter 6. Test Report

Table 6-1. Test report

Test case	Status
Fastboot flash	Pass
QFIL Flat flash	Pass
QFIL Meta flash	Pass
RAM/disk memory	Pass
Recovery mode	Pass
Power on button	Pass
Volume up/down button	Pass
Force USB Boot button	Pass
Fastboot (adb & volume down)	Pass
TURBOX_ID	Pass
BT/Wi-Fi/59/Red/Green/Blue LED	Pass
LSM6DSOTR sensor	Pass
SD Card auto mount/unmount/multiple plug & unplug	Pass
Debug UART	Pass
Type A connection	Pass
Interaction between Type A and Type C	Pass
Type C connection	Pass
DP	Pass
LCD touch panel	Pass
MIPI LCD	Pass
Audio line in	Pass
Audio headphone	Pass
Audio speaker	Pass
BT connection/speed	Pass
Wi-Fi	Pass
Battery	Pass
POE15W power supply	NA
12V DC power supply	Pass
Charge the battery via type-c	Pass
Ethernet_Network port indicator	Pass
DHCP client	Pass
Sleep mode wakeup	Pass
System bootup time	Pass
Random read/write	Pass
2 x IMX577 module capture & video	Pass
2 x OV2735 module capture & video	Pass
2 x IMX577 front and rear camera switchover	Pass
IMX415 module capture & video	Pass
HDMI IN	Pass

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Chapter 7. Known Issues

Table 7-1. List of known/fixed issues

Issue No.	Description	Status
Limitations-001	Incompatible with 2GB DDR.	Open
Limitations-002	RJ45 indicator is abnormal.	Open
Limitations-003	IMX577 module camera preview scale must be 1:1.	Open
BUG-2661	HDMI in compatibility needs to be optimized.	Fixed

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